Linear Algebra II HW 4

Due Date: on or before 18:00, Mar/13/2024 (Thursday), in tutorial.

1. Let U be a subspace of a finite dimensional vector space V. Suppose dim U = dim V. Show that $U = V$.

2. Let A be an $m \times n$ matrix of rank r. Show that

If $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_r\}$ is a basis of $\text{Col}(A^T)$, then $\{A\boldsymbol{v}_1, A\boldsymbol{v}_2, \cdots, A\boldsymbol{v}_r\}$ is a basis of Col(A).

3. Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be linearly independent column vectors in $\boldsymbol{R}^n$ and let $S = \text{Span}(\boldsymbol{x}, \boldsymbol{y})$.We can use $\boldsymbol{x}$ and $\boldsymbol{y}$ to define a matrix A by setting $A = \boldsymbol{x}\boldsymbol{y}^T + \boldsymbol{y}\boldsymbol{x}^T$.

(a) Show that A is symmetric.

(b) Show that $N(A) = S^{\perp}$.

(c) Show that the rank of A must be 2.

4. For each of the following pairs of vectors **x** and **y**, find the vector projection **p** of **x** onto **y** and verify that **p** and **x** − **p** are orthogonal:

(a) $\boldsymbol{x} = (3,5)^T$, $\boldsymbol{y} = (1,1)^T$

(b) $\boldsymbol{x} = (2,-5,4)^T$, $\boldsymbol{y} = (1,2,-1)^T$

5. For each of the following matrices, determine a basis for each of the subspaces $\text{Col}(A^T)$, $N(A)$, $\text{Col}(A)$, and $N(A^T)$:

(a) $A = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 4 & 0 \end{pmatrix}$

(b) $A = \begin{pmatrix} 4 & -2 \\ 1 & 3 \\ 2 & 1 \\ 3 & 4 \end{pmatrix}$

6. Let U and V be a subspace of $\boldsymbol{R}^n$ with usual dot product. Prove the following

(a) If $U \subset V$, then $V^{\perp} \subset U^{\perp}$.

(b) $(U+V)^{\perp} = U^{\perp} \cap V^{\perp}$. Hence $(U \cap V)^{\perp} = U^{\perp} + V^{\perp}$.